



CZ/CE2002/SC2002: Object-Oriented Design & Programming

Introduction (Second Half)

Dr. Li Fang

College of Computing and Data Science



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Contact Information

Dr. Li fang

Senior Lecturer, College of Computing and Data Science

Contact Information:

- Office: N4-02B-42
- Phone: (+65) 6790 6106
- Email: asfli@ntu.edu.sg

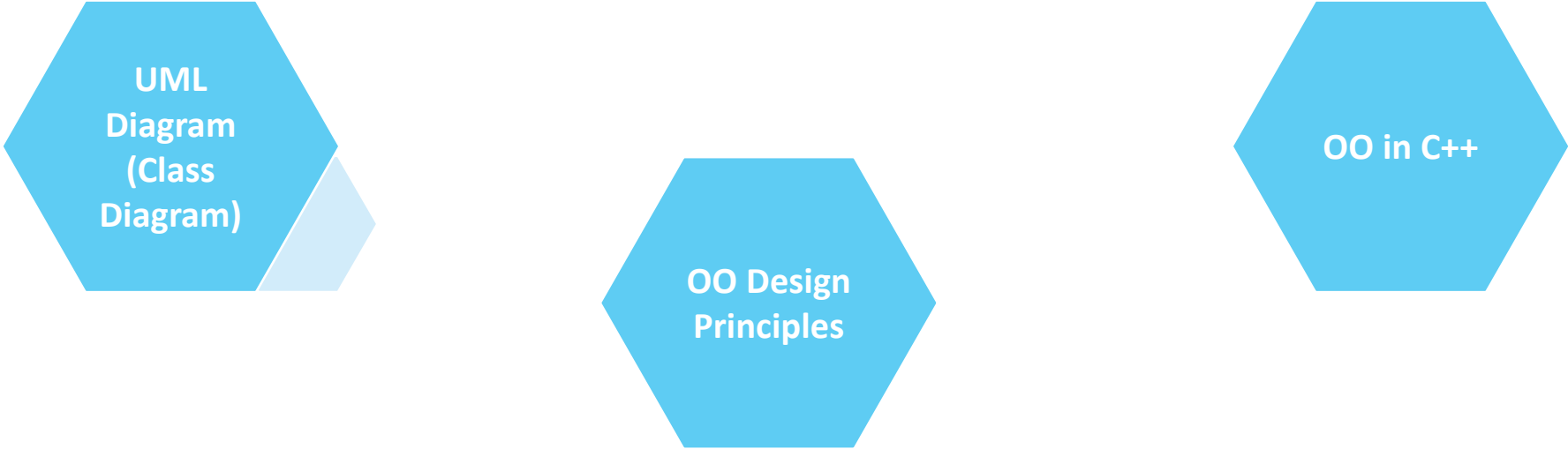
Consultation Hours:

- Send an email for an appointment



Course Outline

The following topics will be covered during the second half of the course:



UML
Diagram
(Class
Diagram)

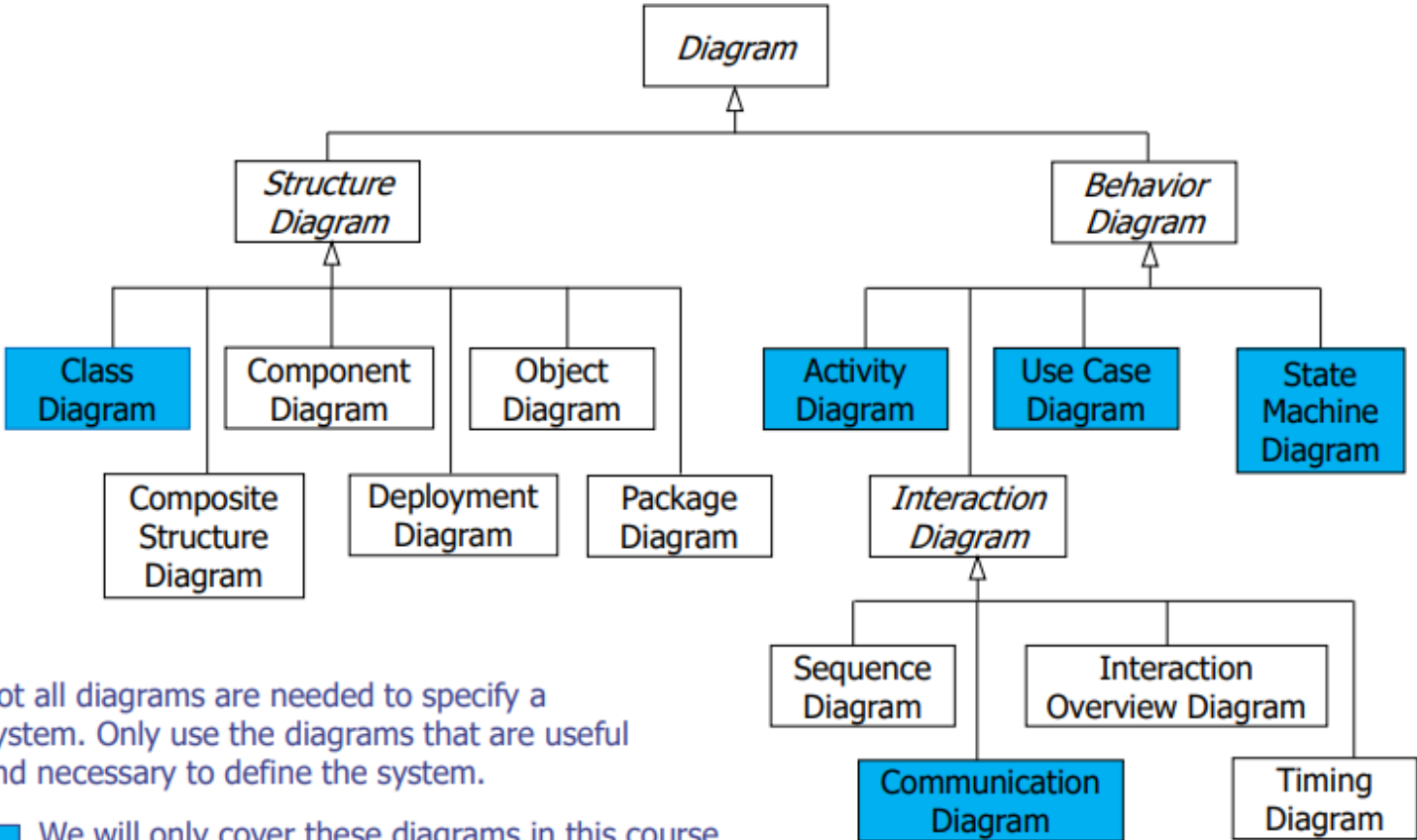
OO Design
Principles

OO in C++

CZ 3002 – Advanced Software Engineering

Course Code	CZ3002
Course Title	Advanced Software Engineering
Pre-requisites	CZ/CE2006: Software Engineering

UML Diagram Types



Not all diagrams are needed to specify a System. Only use the diagrams that are useful and necessary to define the system.

 We will only cover these diagrams in this course (Others will be covered in CZ3003)

Softwares used in Second half

Attached Files:  Importing VP UML License Key.pdf (269.219 KB)

In lecture, the following softwares will be used in demo.

- Eclipse for Java programming. (Same as first half)
- Code::Blocks for C++ programming. (You have learnt it in CX1103/SC1003)
- Visual Paradigm for UML Diagram drawing. (You might refer to the following 10 min's demo if you are new to Visual Paradigm)

◦ **license key:**

The license key for UML 17.2 Standard Edition is 3C543-J6RRJ-7KNL7-QNMUQ-QY748

, which is free for CCDS students since CCDS has purchased academic license for all CCDS

students..

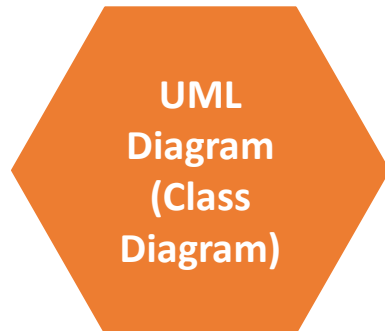
You need download version 17.2 and use your NTU email to register for the verification code.

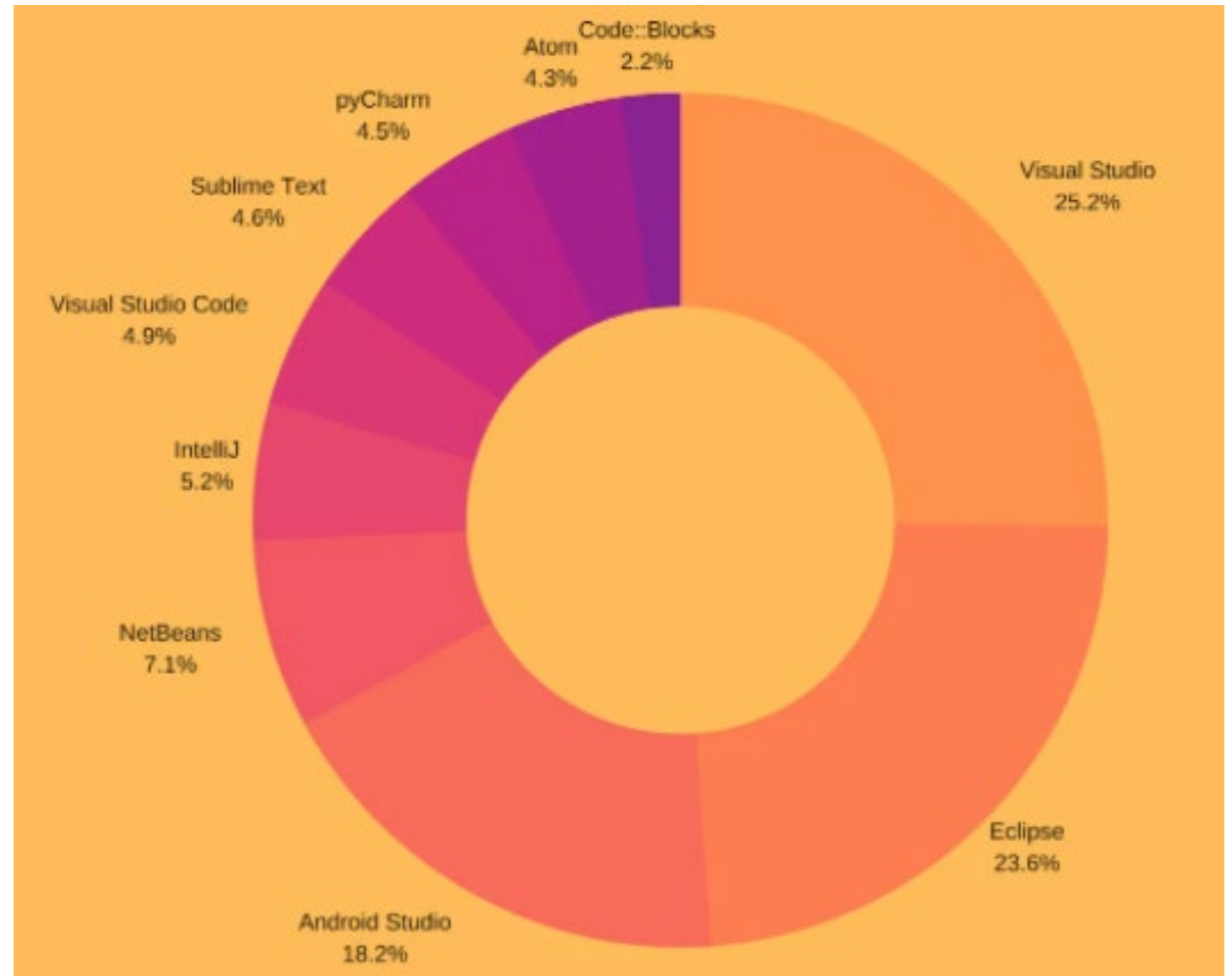
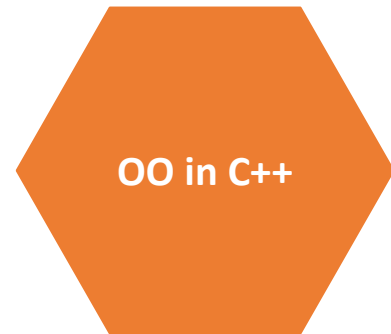
- <https://www.youtube.com/watch?v=O2Qlw1QmCWQ>

You are free to use you own preferred software.



Visual Paradigm





~~A guide to pace your learning of online lecture contents.~~

Blended Approach:

(1) TEL video – first cut understanding of topics covered in short videos.

Link -> NTULearn SC/CE/CZ2002 Lecture site > Content > SC/CE/CZ2002 Object Oriented Design & Programming (Second Half)

(2) Review session with MCQ questions, Key/difficult points explanation, sample diagram/code demo and Q&A - targeted revision questions to apply and recap concepts learned, clarify your doubts via Q&A.

Below is a guide to pace your learning prior to attending the Review sessions on Tuesday and Friday. If you can view more or ahead, you are encouraged to do so – example to headstart your project assignment.

In the TEL video, you can enable the English subtitle by clicking on 'cc' at the bottom right corner. There are also time point in the slides and sub-sections of topics for your easy reference to the contents in the video.

TEL Video under Content folder			Face-to-face sessions (Tuesday and Friday)	Tutorials	
Week 8		Introduction: Second Half Week 8: UML Model Class Relationships - Class Diagram (Object Diagram, Class Stereotypes: not examinable)	Week 8-week 11: Two revision lectures per week. Tuesday and Friday. MCQ questions, Key/difficult points explanation, sample diagram/code demo and Q&A - targeted revision questions to apply and recap concepts learned, clarify your doubts via Q&A.	Tutorial 5: Inheritance & Polymorphism	
Week 9		Week 9: Design Principles		Tutorial 6: Class Diagram	
Week 10				Week 10: OO Concepts in C++ part 1	Tutorial 7: (e-learning) Design Principles
Week 11			Week 11: OO Concepts in C++ part 2		Tutorial 8: Object-Oriented in C++
Week 12		Week 12: -Review of students' questions -quiz revision			Tutorial 9: Polymorphism in C++
Week 13,14				NO classes. Assignment work. Assignment Consultation, if necessary. SC2002 Online Quiz Week14 – Tuesday Physical online quiz in labs Week 14 Friday: Group Assignment code and doc submission, oral presentation, deadline	

Course Assessment Components

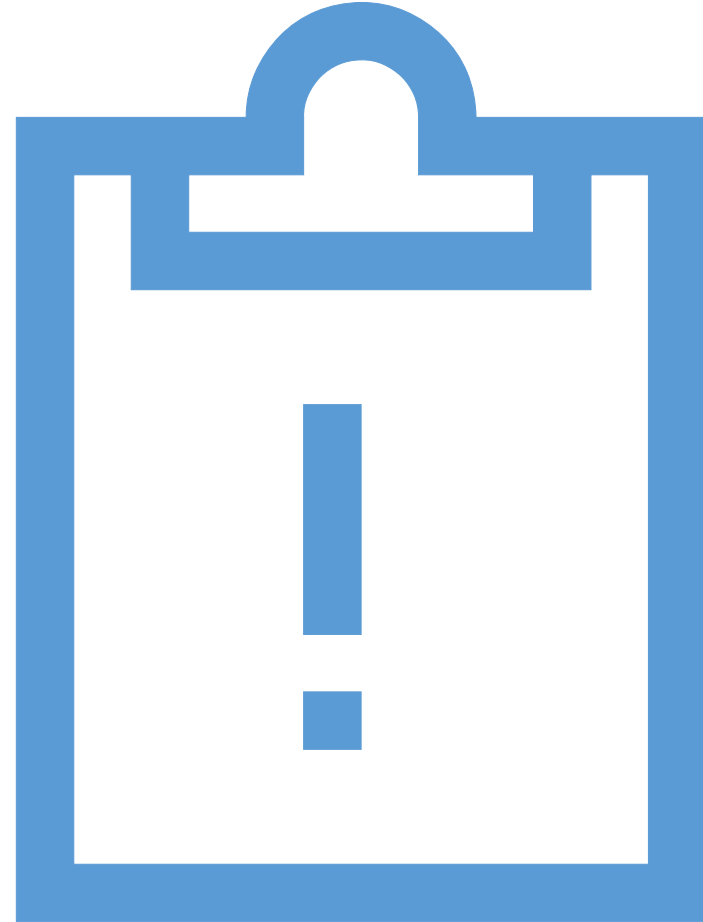
- 5% Online quiz on Basic Java Syntax
- 45% Two Physical Quizzes
 - One in Week 7 (Done)
 - One in Week 14 (Tuesday November 19)
- 10% Lab completion
- 40% Group assignment
- No Final Exam!

Quiz 2: Online MCQ quiz

- **Date: Week 14 (Tuesday November 19)**
- **Time: AM session: 9:30-10:45. PM session: 2:00-3:15.**
- **Duration: 1.25 hours (75 Mins).**
- **Question type: MCQ. Single (default)/Multiple correct (state specially) answers MCQs.**
- **Content: All lecture content.**
- **Number of questions: 37.**
- **Venue: CCDS labs.**
- **Closed book. Lockdown browser is required.**

Group Assignment Assessment components

- UML Class Diagram
- Design Consideration
- Implementation Code
- Demonstration and report



Group Assignment 40%

Deadline: **Friday of Week 14**
November 22 2024

- Group Assignment code, class diagram, and report submission deadline: **One day before your scheduled oral with your TA**
- Group Assignment oral presentation deadline
 - Via Physical/Online meeting
 - Arrange Group Assignment oral presentation date and time earlier with your TA
 - Every student must take turn to present



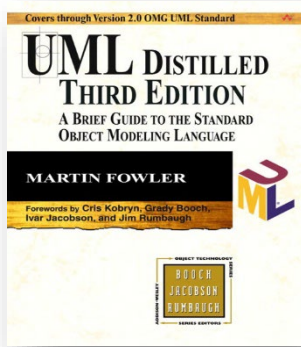
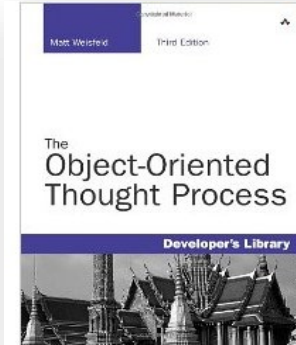
Group Assignment Briefing Q&A

Friday of Week 9 (Oct. 18)

Course Reference Books



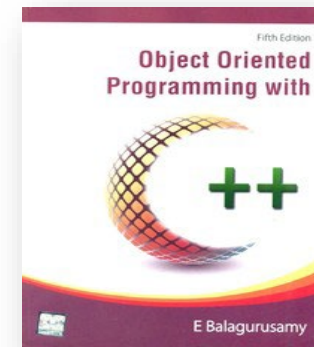
The Object-Oriented Thought Process:
Matt Weisfeld, 4th Edition, Pearson / Addison
Wesley



UML Distilled:
A Brief Guide to the Standard Object Modeling
Language, 3rd Edition, Martin Fowler, Pearson /
Addison Wesley



Object Oriented Programming With C++:
Fifth Edition, E Balagurusamy



Thank you!